

FABRICATION NOTES

DRC'S MUST BE RUN ON THE GERBERS PRIOR TO BUILDING BOARDS

1.) BOARD MATERIAL: (SEE STACKUP BELOW)

- RF35    □ R04003  
■ FR-4    □ R04350  
□ 4000-13EP

2.) BOARD STACKUP

- COMPONENTS ON TOP SIDE ONLY    □ COMPONENTS ON BOTH SIDES

LAYERS

FINISHED OVERALL THICKNESS:

- .031 +/- .005  
■ .062 +/- .005  
□ .093 +/- .005  
□ .125 +/- .005

3.) COPPER WEIGHT:

EXTERNAL LAYERS: (SEE STACKUP BELOW) OUNCE FINISHED

INTERNAL LAYERS: (SEE STACKUP BELOW) OUNCE FINISHED

4.) PLATING:

- HOT AIR SOLDER LEVEL (HASL) EXPOSED AREAS WITH LEADED SOLDER  
FINISH 50uIN MIN - 200uIN MAX.  
■ FINISH: AS PER IPC ENIG. SPECIFICATION 4522, 2-4 MICRO-INCHES  
IMMERSION GOLD OVER 120-240 MICRO-INCHES NICKEL  
□ OMIKRON WHITE TIN DEPOSITION PROCESS OR ANDREW APPROVED EQUIVALENT  
TIN PLATING SHALL BE 200 TO 250 MICRO INCHES IN THICKNESS.  
□ STANDARD SPEC.

5.) DRILLING:

UNLESS OTHERWISE SPECIFIED, PLATE ALL HOLES THRU.  
SEE HOLE CHART FOR FINISHED HOLE SIZES AND QUANTITY.

6.) HOLE SIZE TOLERANCES SHALL BE:

(SEE HOLE CHART)

7.) SOLDERMASK:

- TOP SIDE  
■ BOTTOM SIDE  
■ LIQUID PHOTO-IMAGEABLE SOLDERMASK SHALL BE APPLIED  
OVER BARE COPPER UNLESS OTHERWISE SPECIFIED  
■ GREEN

- BLUE  
□ RED  
□ BLACK  
□ MATTE BLACK

8.) SILKSCREEN:

- TOP SIDE  
■ BOTTOM SIDE  
■ CHARACTER LEGEND TO BE SCREENED USING NON-CONDUCTIVE  
EPOXY BASED INK.  
■ WHITE  
□ \_\_\_\_\_

9.) BOW AND TWIST: MAXIMUM ACCEPTABLE IS 0.75% (IPC-TM-650)

10.) FABRICATE IN ACCORDANCE WITH IPC-6011/6012, CLASS 2 (LATEST REVISIONS)

11.) ACCEPTABILITY TO IPC-A-600 (LATEST REVISIONS)

12.) CONDUCTOR INFORMATION:

DRC MUST BE RUN ON GERBERS BEFORE BUILDING BOARDS UNLESS

APPROVAL IS GIVEN IN WRITING.

SMALLEST TRACE: 6 mil

SMALLEST SPACING: 5.91 mil

CONDUCTOR WIDTHS AND SPACING SHALL BE WITHIN +/-20% OF ARTWORK ORIGINALS.

CONTROLLED IMPEDANCE REQUIREMENTS AS CALLED OUT IN CONTROLLED IMPEDANCE NOTES.

13.) REMOVE ALL BURRS AND BREAK SHARP EDGES 0.015 INCHES MAX

14.) VENDOR INFORMATION:

VENDOR WILL PUT THEIR UL LOGO AND DATE CODE IN SILKSCREEN

ON THE COMPONENT SIDE IN THE UNOBSTRUCTED AREA

15.) ALL 0.300mm VIA-IN-PAD SHALL BE PLATED, FILLED, AND FLATTENED.

BOARD CHARACTERISTICS

Copper Layer Count: 6    Board Thickness: 60.87 mils

Board overall dimensions: 9500.00 mils x 3000.00 mils

Min track/spacing: 6.00 mils / 5.91 mils    Min hole diameter: 11.81 mils

Copper Finish: ENIG    Impedance Control: Yes

Castellated pads: No    Plated Board Edge: No

Edge card connectors: No

| Layer Name | Type                | Material      | Thickness (mils) | Color         | Epsilon R | Loss Tangent |
|------------|---------------------|---------------|------------------|---------------|-----------|--------------|
| SST        | Top Silk Screen     | Not specified | 0 mils           | Not specified | 1         | 0            |
| PMT        | Top Solder Paste    |               | 0 mils           |               | 1         | 0            |
| SMT        | Top Solder Mask     | Not specified | 0.3937 mils      | Green         | 3.3       | 0            |
| LY01       | copper              |               | 1.37795 mils     |               | 1         | 0            |
| Dielectric | prepreg             | 2116          | 5.11811 mils     | FR4 natural   | 4,45      | 0            |
| LY02       | copper              |               | 1.37795 mils     |               | 1         | 0            |
| Dielectric | core                | DK            | 16.92913 mils    | FR4 natural   | 4,6       | 0            |
| LY03       | copper              |               | 1.37795 mils     |               | 1         | 0            |
| Dielectric | prepreg             | 7628          | 7.71654 mils     | FR4 natural   | 4,74      | 0            |
| LY04       | copper              |               | 1.37795 mils     |               | 1         | 0            |
| Dielectric | core                | DK            | 16.92913 mils    | FR4 natural   | 4,6       | 0            |
| LY05       | copper              |               | 1.37795 mils     |               | 1         | 0            |
| Dielectric | prepreg             | 2116          | 5.11811 mils     | FR4 natural   | 4,5       | 0.02         |
| LY06       | copper              |               | 1.37795 mils     |               | 1         | 0            |
| SMB        | Bottom Solder Mask  | Not specified | 0.3937 mils      | Green         | 3.3       | 0            |
| PMB        | Bottom Solder Paste |               | 0 mils           |               | 1         | 0            |
| SSB        | Bottom Silk Screen  | Not specified | 0 mils           | Not specified | 1         | 0            |

Controlled Impedance Top and Bottom Layers

1 Top Sig Edge coupled microstrip .17/.15 mm 100 Ohm differential 10%

2 In1 GND

3 In2 PWR

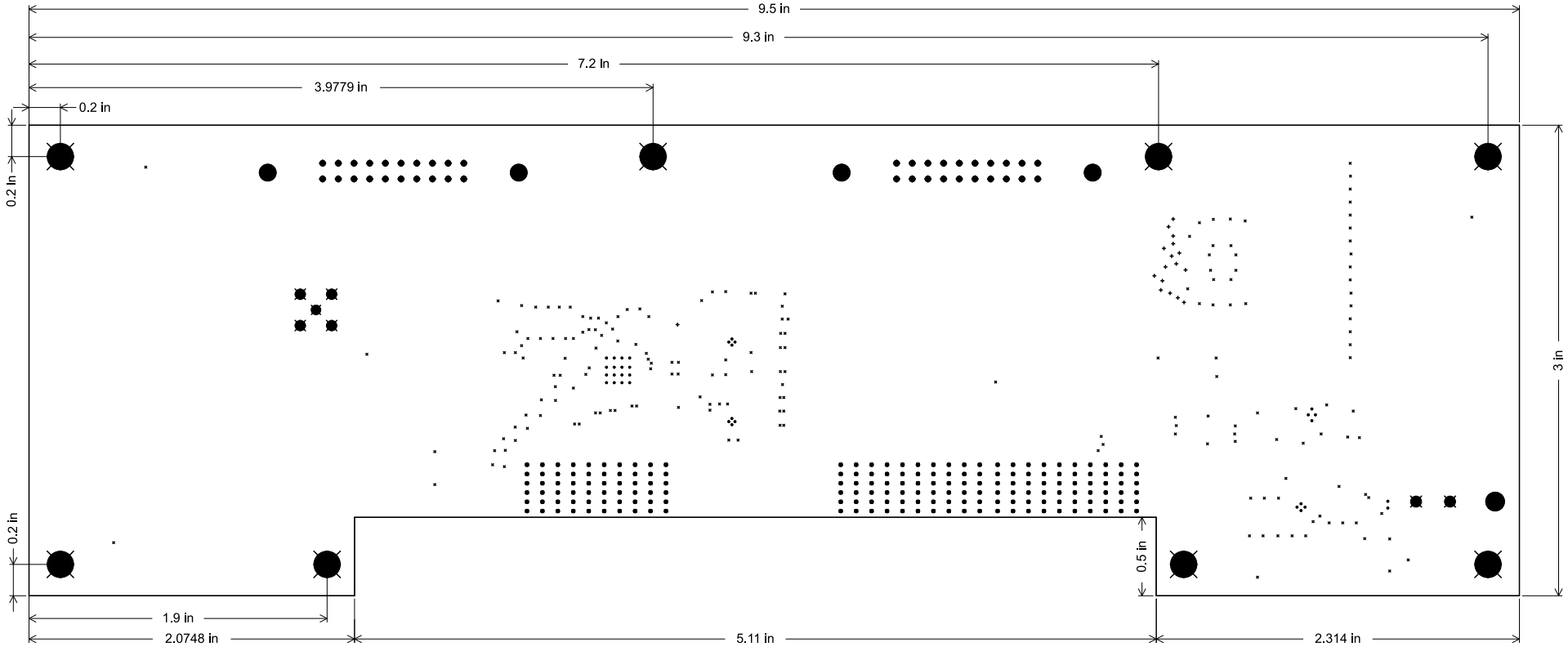
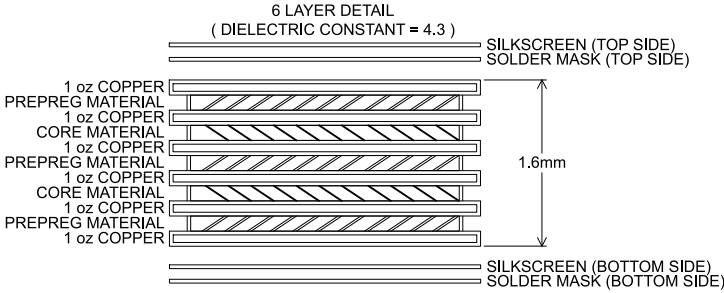
4 In3 Sig

5 In4 GND

6 Bot Sig

Edge coupled microstrip .17/.15 mm 100 Ohm differential 10%

THESE NUMBERS ARE FOR REFERENCE ONLY. FINAL WIDTHS AND SEPERATIONS TO ACHIEVE  
100 OHMS DIFFERENTIAL IMPEDANCE SHALL BE THE RESPONSIBILITY OF THE BOARD FABRICATOR.



DRILL DRAWING  
SK-2601-52  
REV-1

Drill Map:

- 0.300mm / 0.0118" (190 holes)
- 0.300mm / 0.0118" (34 holes)
- 0.500mm / 0.0197" (17 holes)
- 0.510mm / 0.0240" (180 holes)
- 0.970mm / 0.0382" (40 holes)
- ⊗ 1.500mm / 0.0591" (1 hole)
- ⊗ 1.700mm / 0.0669" (4 holes)
- ⊗ 1.800mm / 0.0709" (2 holes)
- ⊗ 4.300mm / 0.1693" (8 holes)
- ⊗ 2.690mm / 0.1059" (4 holes) (not plated)
- 3.000mm / 0.1181" (1 hole) (not plated)

ALL HOLES 0.610MM / 0.0240" PRESS FIT TO -0.84 mil / +1.68 mil

UNLESS OTHERWISE SPECIFIED  
ALL DIMENSIONS ARE IN INCHES  
DIMENSIONS IN BRACKETS [XXXX] (WHERE PRESENT)  
ARE MILLIMETERS AND ARE FOR REFERENCE ONLY  
INTERPRET DRAWING AS PER ANSI Y14.5-1994 OR Y14.5-1995

|                                                                                                 |                                              |
|-------------------------------------------------------------------------------------------------|----------------------------------------------|
| DIMENSIONAL TOLERANCES<br>X = +/- 0.060<br>X = +/- 0.030<br>.XX = +/- 0.015<br>.XXX = +/- 0.005 | ANGULAR TOLERANCE<br>±0.5°<br>FINISH<br>125/ |
|-------------------------------------------------------------------------------------------------|----------------------------------------------|



BROOKHAVEN NATIONAL LABORATORY  
BROOKHAVEN SCIENCE ASSOCIATES  
UPTON, NEW YORK 11973  
Exploring Life's Mysteries,  
Protecting The Future



PCB FAB DRAWING  
ACM-II AFE Board

|               |                |     |
|---------------|----------------|-----|
| CHECKED BY    | Joe Mead       |     |
| APPROVAL      | N/A            | N/A |
| ENGINEER      | Joe Mead       |     |
| APPROVAL      | Danny Padrazo  |     |
| SUPERVISOR    |                |     |
| APPROVAL      | Bob Lee        |     |
| ES&H APPROVAL | Nicole Magbulo |     |

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SK-2601-52

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NEXT ASSY:

PROJECT:



THIRD ANGLE  
PROJECTION

SCALE:  
SEE DWG VIEW

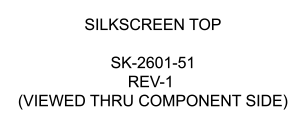
WBS#

SHEET SIZE

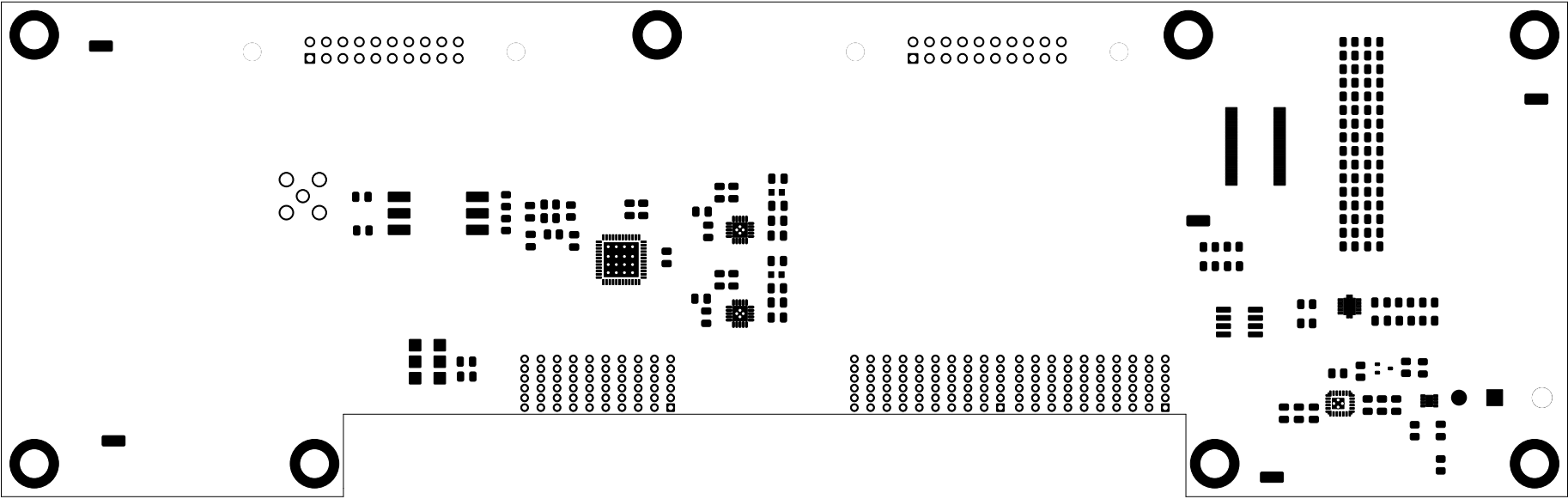
ESHQ  
REQ LEVEL

DRAWING/PART NUMBER

SHEET 1 OF 1

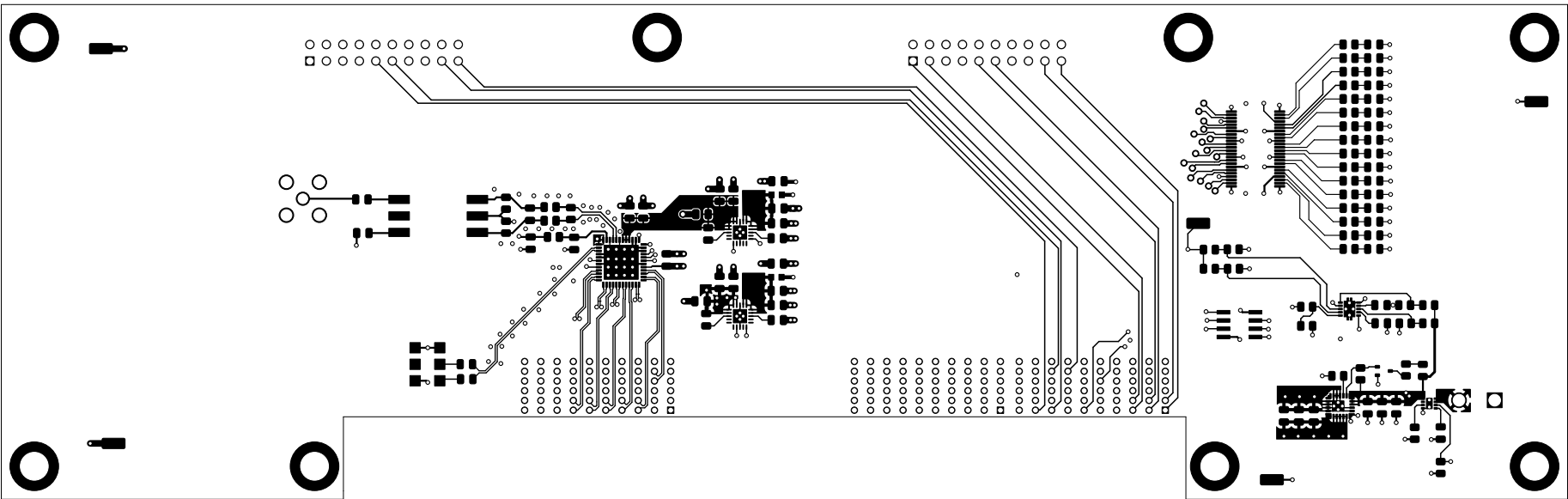


SILKSCREEN TOP  
SK-2601-51  
REV-1  
(VIEWED THRU COMPONENT SIDE)

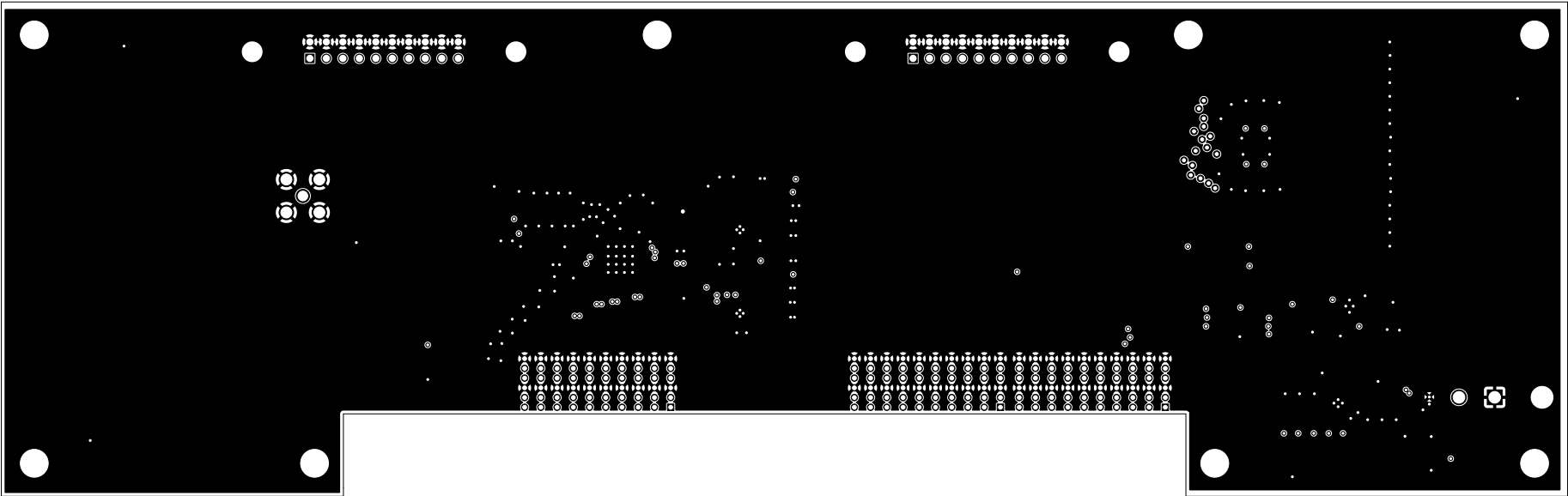


SOLDER MASK TOP  
SK-2601-51  
REV-1

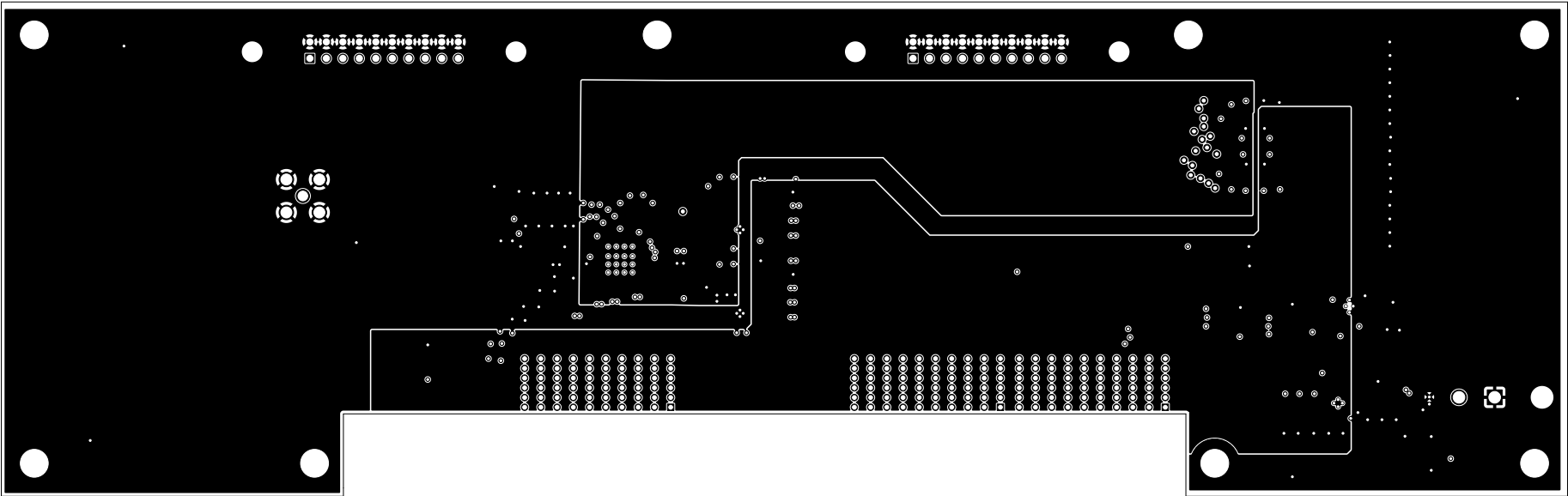




LAYER 1 COMPONENT SIDE  
SK-2601-51  
REV-1

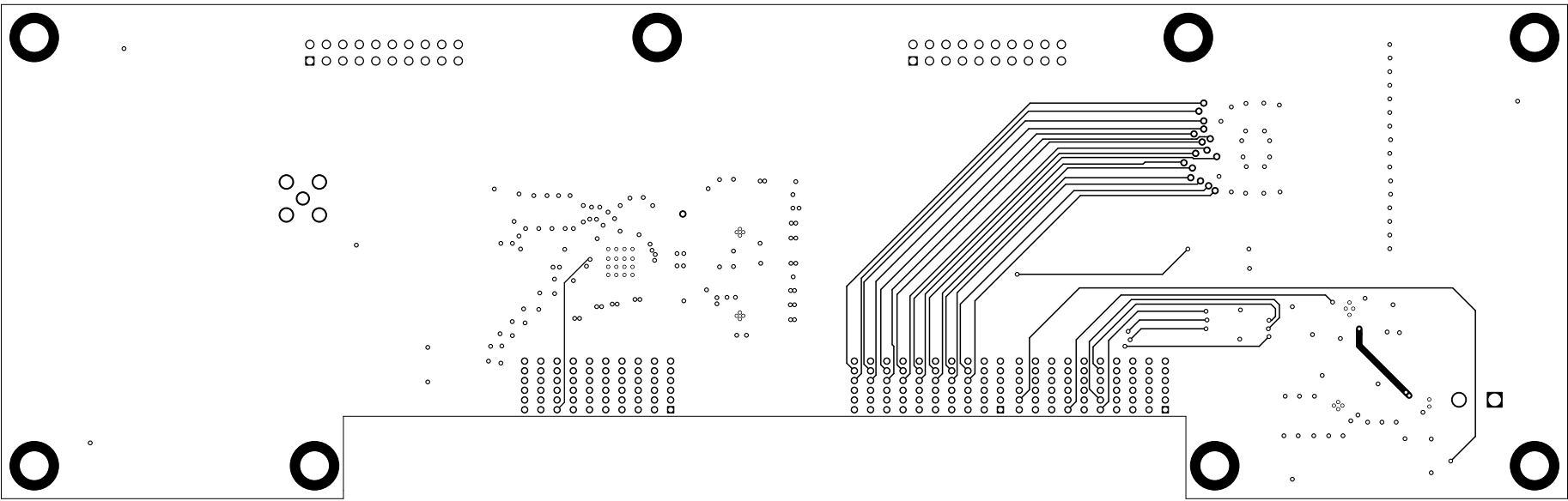


LAYER 2 INTERNAL PLANE  
SK-2601-51  
REV-1

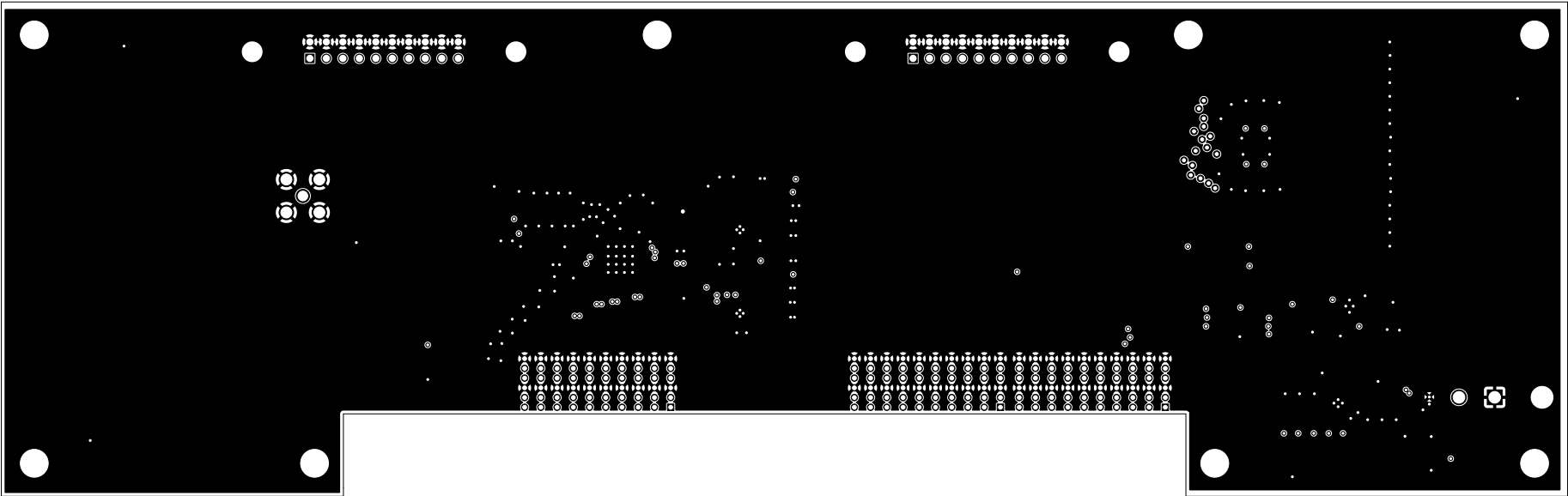


LAYER 3 INTERNAL PLANE  
SK-2601-51  
REV-1



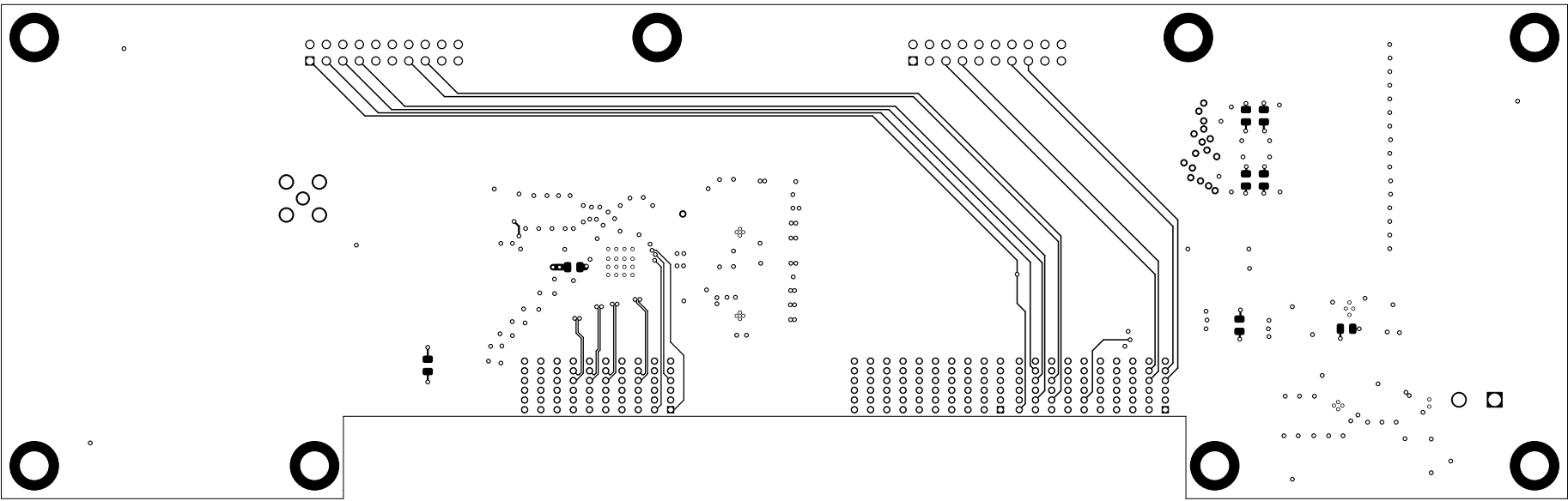


LAYER 4 INTERNAL SIGNAL  
SK-2601-51  
REV-1

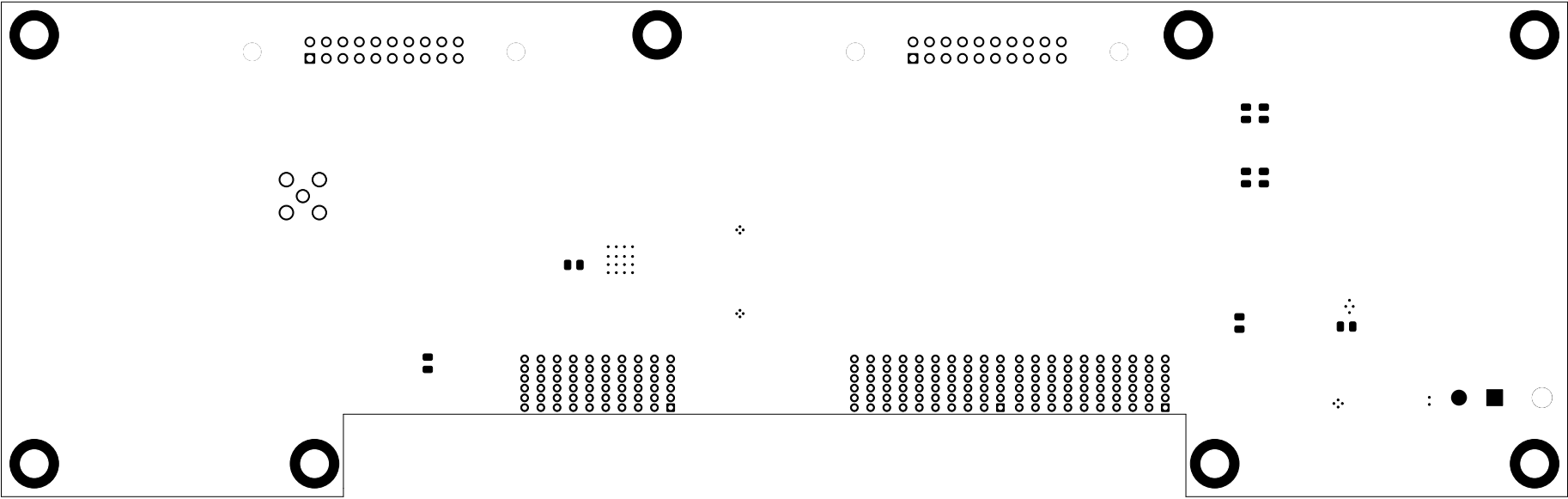


LAYER 5 INTERNAL PLANE  
SK-2601-51  
REV-1

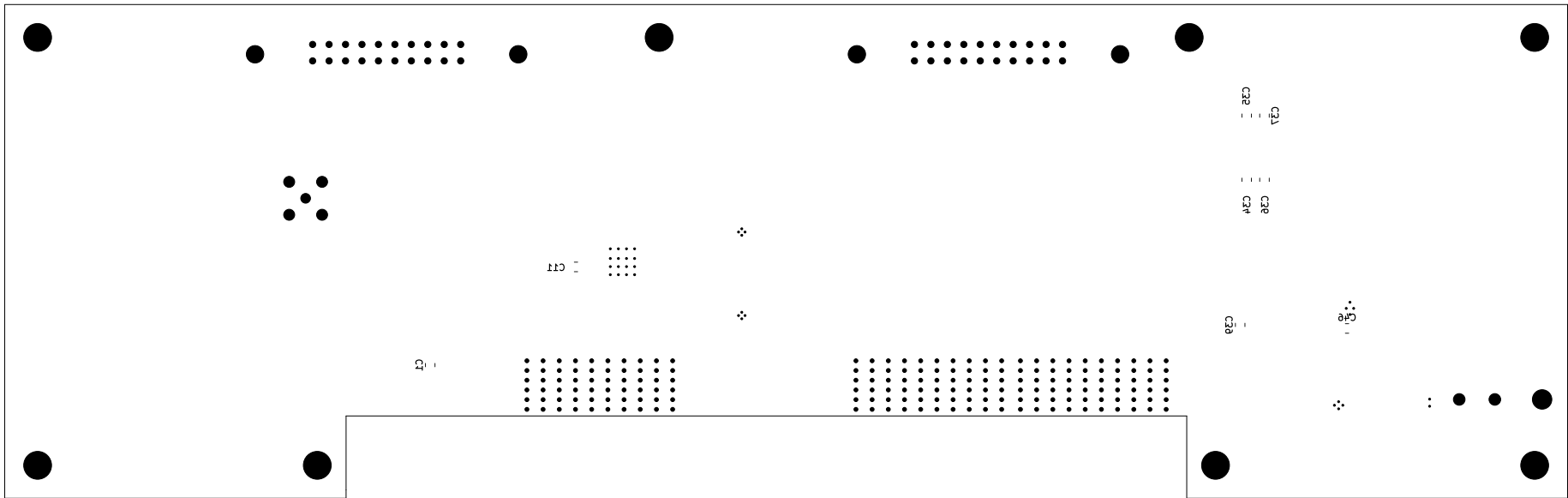




LAYER 6 BOTTOM SIDE  
SK-2601-51  
REV-1



SOLDER MASK BOTTOM  
SK-2601-51  
REV-1



REV-1  
8K-5801-81  
SILKSCREEN BOTTOM